HACKATHON TRACK: REINVENT COMMERCE · ACT III

PayLens Deterministic Cross-Rail Revenue Reconciliation

Autonomous Financial Deal Desk for Fragmented E-Commerce Platforms

Live on Streamlit Cloud · AMD Radeon Cloud TokenFactory & Google Gemma 4 REST

PILLAR 01

TRACK 3: REINVENT COMMERCE

Autonomous Deal Desk for Multi-Rail E-Commerce

PILLAR 02

LIVE HARDWARE: AMD RADEON CLOUD

DeepSeek-V4-Flash on TokenFactory + MI300X vLLM Recipe

PILLAR 03

AI REASONING: DUAL-ENGINE INFERENCE

DeepSeek-V4-Flash & Google Gemma 4 with 1-Click Action Dispatch

PILLAR 04

DETERMINISTIC VERIFICATION

DuckDB Math Matchers + Holt-Winters + 100% Regex Shield

CORE OPERATING PHILOSOPHY

Code Computes Every Dollar. AMD AI Explains and Recovers.

No LLM floating-point calculations. Zero hallucinated currency figures. All numbers validated against DuckDB ledger.

GITHUB REPOSITORY

https://github.com/labib-morol/paylens-amd

LIVE WEB APP

Streamlit Cloud (Global Deployment)

AUDIT STATUS

3-Gate Automated Verification 100% Passed

MARKET REALITY & OPERATIONAL RISK

The Multi-Rail Reconciliation Problem

Modern E-Commerce Operates on Disconnected, Asynchronous Financial Rails

PILLAR 01

1. DIGITAL STOREFRONTS

Shopify · WooCommerce · Magento

- Asynchronous order dispatch webhooks
- Volatile order status flags (Fulfilled vs Pending)
- Unrecorded cancellations & split-shipment leaks
- High-velocity catalog updates hide silent drops

PILLAR 02

2. PAYMENT GATEWAYS

Stripe · PayPal · bKash · SSLCommerz

- Net payouts obscure individual order deductions
- Unrecorded processing fee tiers & rolling reserves
- 3-day settlement delays create matching drift
- Silent authorization failures drop customer funds

PILLAR 03

3. CARRIER REMITTANCES

DHL · FedEx · ExpressBD COD

- Cash-on-Delivery (COD) cash collected in the field
- Bundled PDF remittance statements with deductions
- Return surcharges & lost transit shortfalls
- Spreadsheet-based reconciliations happen weeks late

REAL-WORLD AUDIT IMPACT

$10,975.00

Silent Revenue Leakage Detected across 20 Synthetic Storefronts

WHERE THE MONEY SILENTLY LEAKS

• **Dispatched Unpaid Orders:**	$1,255.00	(Orders delivered with zero gateway authorization record)
• **Carrier Remittance Shortfalls:**	$175.00	(Under-settled COD remittances hidden in carrier PDF statements)
• **Stockout Margin Exposure:**	$1,597.50	(Hero SKU stockouts projected via Holt-Winters forecast)

ARCHITECTURAL FOUNDATION

Core Philosophy: Code Computes. Model Explains.

Strict Separation of Mathematical Truth and Language Model Reasoning

STAGE 01 · MATHEMATICAL GROUND TRUTH

LAYER 1: DETERMINISTIC ENGINE (DuckDB + statsmodels)

- 100% of arithmetic calculations, relational joins, and statistical demand forecasts run in compiled code.
- DuckDB exact order matching + 30-minute fuzzy reconciliation joins between orders and payment webhooks.
- statsmodels Holt-Winters triple exponential smoothing forecasts stockout margin exposure deterministically.
- Zero floating-point hallucination. Zero probabilistic math. Output is verified ledger truth.

STAGE 02 · STRICT NUMERIC DEFENSE

LAYER 2: NUMERIC GROUNDING SHIELD (Regex Guardrail)

- Every single generated number and currency token must come verbatim from ALLOWED_NUMBERS evidence list.
- Validates dollar figures against DuckDB source rows (e.g. $1,255.00, $175.00, $1,597.50, quantities).
- If an unverified token appears, automatically triggers a 1-turn correction retry citing the ungrounded token.
- If correction fails, instantly reverts to deterministic template fallback to guarantee 100% financial integrity.

STAGE 03 · EXPLANATION & OUTREACH

LAYER 3: REASONING LAYER (AMD TokenFactory & Google AI Studio)

- AMD Radeon Cloud TokenFactory (DeepSeek-V4-Flash) and Google Gemma 4 REST API (gemma-4-31b-it).
- Translates complex multi-rail ledger discrepancies into actionable, audit-ready plain English summaries.
- Auto-drafts dispute messages tailored to merchant tone: Polite Reminder, VIP 5% Discount, Carrier Dispute.
- Model only explains and drafts. It never computes.

Rule 1: LLMs explain discrepancies and draft correspondence. Code calculates every dollar. Zero exceptions.

ALGORITHMIC PRECISION

Deterministic Detectors & Mathematical Formulations

Four Specialized Engines Covering Every Dimension of Commerce Risk

PAYMENT LEAKAGE DETECTOR

1. Dispatched Unpaid Matcher

Methodology: 2-Pass Exact + 30-Minute Fuzzy Relational Join

```
dispatched_orders.order_id = rail_payments.order_id
OR (|order_created_at - payment_created_at| <= 30m)
```

Operational Logic:
Dispatched orders with zero payment authorization record within matching window.

Audited Evidence:
Found $1,255.00 unpaid dispatched order (ORD-7819) at store merchant_441.

LOGISTICS AUDIT ENGINE

2. Carrier Remittance Shortfall

Methodology: Bundled PDF Parser & Net Remittance Balance Formulation

```
Expected Net = cod_collected - courier_fee - surcharge
Shortfall = Expected Net - actual_settled
```

Operational Logic:
Extracts consignment numbers from carrier settlement statements and joins to ledger.

Audited Evidence:
Detected $175.00 carrier under-settlement across consignments in ExpressBD PDF.

INVENTORY RISK PREDICTOR

3. Holt-Winters Stockout Forecaster

Methodology: Triple Additive Exponential Smoothing Demand Formulation

```
Days of Cover = current_stock / daily_forecast
Lost Margin = (Lead Time - Days of Cover) * daily_forecast * unit_margin
```

Operational Logic:
Projects stockout window and lost profit margin over supplier restocking lead time.

Audited Evidence:
Prioritizes Hero Blazer SKU-APX-01 ($1,597.50 margin at risk) under stockout objective.

MACRO VOLUME SURVEILLANCE

4. STL Seasonal Residual Anomaly Engine

Methodology: LOESS Seasonal-Trend Decomposition Formulation

```
Y_t = Trend_t + Seasonal_t + Residual_t
Flag if |z_score(Residual_t)| > 2.2
```

Operational Logic:
Separates true revenue anomalies from recurring weekly/monthly seasonal fluctuations.

Audited Evidence:
Provides merchant volume baseline context to prevent false operational alarms.

EMPIRICAL VALIDATION

Ground Truth & Stress-Test Evaluation

Audited Evaluation on Adversarial Held-Out Synthetic Datasets (F1: 0.933)

HELD-OUT STRESS-TEST BENCHMARK

30 Planted Anomaly Records

F1 SCORE	PRECISION
0.933	0.933

RECALL	TRUE POSITIVES
0.933	28 / 30

AUDITED CONFUSION MATRIX BREAKDOWN

Classification	Count	Audit Description
True Positives (TP)	28	Confirmed payment leakages and genuine stockouts captured
False Positives (FP)	2	Late settlements flagged for human operator review (e.g. SEPA)
False Negatives (FN)	2	Complex multi-split consignments under fuzzy threshold
True Negatives (TN)	170	Clean settled orders correctly confirmed without alert spam

Stress test generated with seed 1337 and adversarial noise injection.

DOCUMENTED EDGE CASE & AUDIT TRANSPARENCY

Transparent Disclosure: 48h SEPA Wire Delay

SCENARIO: ORD-FAIL-4303 ($850.00 European Wire)

A legitimate business customer in Germany sent a SEPA bank wire of $850.00.

The merchant dispatched immediately, but European interbank clearing took 48 hours.

Settled 48h later (exceeding 30-min window), so the detector flagged it as leakage.

Outcome: Transparently documented as 1 of the 2 False Positives in our stress-test split.

WHY TRANSPARENCY WINS IN ENTERPRISE FINANCE

- **No Fabricated 100% Accuracy:** Competitors claim 100% precision by hiding delayed wires under the rug. PayLens transparently exposes the edge cases.
- **1-Click Human Triage:** In the Deal Desk Inspector, operators can review the wire transfer timestamp and clear the flag in a single click.
- **Immutable Audit Trail:** All decisions create immutable SHA-256 receipts logged with timestamp, user ID, and model grounding trace.

PERFORMANCE & PROVENANCE

Authentic Hardware Telemetry & Scaling

Measured Dual-Engine Benchmarks on AMD Radeon Cloud & Google AI Studio

HARDWARE: AMD RADEON CLOUD TOKENFACTORY

DeepSeek-V4-Flash Serving Telemetry

Real live concurrency requests measured against AMD cloud endpoints

Concurrency Tier 1 (Sequential)

7.22s TTFT · 23.02s Latency

16.8 tok/s

THROUGHPUT

Concurrency Tier 4 (Concurrent Burst)

6.97s TTFT · 24.08s Latency

45.5 tok/s

THROUGHPUT

Concurrency Tier 8 (High Concurrency)

8.32s TTFT · 20.39s Latency

62.9 tok/s

THROUGHPUT

100% FIRST-PASS MODEL GROUNDING PASS RATE

16 of 16 live requests passed numeric regex validation on first pass with zero hallucinations and zero template fallbacks.

DUAL-ENGINE ARCHITECTURE & ON-PREM

Google Gemma 4 & AMD MI300X

GOOGLE GEMMA 4 REST API BENCHMARK

Model: gemma-4-31b-it · Direct AI Studio Integration

COMPLETED	FIRST-PASS GROUNDING	FALLBACKS	ERRORS
2 of 2 (100%)	100% Pass	0 (0.0%)	0

DEDICATED AMD INSTINCT MI300X DEPLOYMENT

Self-Hosting Recipe provided in paylens-amd for enterprise air-gapped finance:

```
docker run --device=/dev/kfd --device=/dev/dri \
  -v /models:/models -p 8000:8000 \
  vllm/vllm-openai:rocm6.2_mi300x \
  --model /models/DeepSeek-V4-Flash \
  --tensor-parallel-size 1 \
  --max-model-len 8192
```

OPERATIONAL DELIVERY & VISION

Human-in-the-Loop Deal Desk & Global Roadmap

Empowering Finance Teams from Local Pilots to Global Scale

PRODUCTION DEAL DESK CAPABILITIES

Autonomous Triage & 1-Click Action

Simulated One-Click Action Approval

Operators approve actions with immutable SHA-256 receipts logged to audit trail.

Multi-Channel Dispute Receipts

Live WhatsApp and Email dispute receipts with verified DuckDB evidence attachments.

Multi-Tone Action Generation

Auto-drafts Polite Reminders, VIP 5% Instant-Pay Discounts, Carrier Disputes, or Expedited POs.

Dynamic Objective Steering

Re-ranks queue on the fly: switch from Urgent Cash Recovery to Prevent Stockouts in 1 click.

GLOBAL MULTI-RAIL EXPANSION

Architected for Global Commerce Rails

Emerging Market Instant Rails — `Brazil Pix, India UPI, Kenya M-Pesa`

Connecting sub-second settlement ledgers to real-time e-commerce dispatch queues.

Real-Time Streaming Engine — `Kafka + ClickHouse Architecture`

Replacing nightly batch jobs with microsecond continuous reconciliation.

Enterprise ERP Connectors — `SAP S/4HANA · Oracle NetSuite`

Direct two-way accounting sync for enterprise treasury and ledger closing.

Autonomous Carrier Dispute Filing — `Carrier Webhooks & Automated Claims`

Filing courier shortfall claims automatically with attached PDF evidence proofs.

ENTERPRISE FINANCE MANIFESTO

Deterministic arithmetic. Grounded AMD intelligence. Built for enterprise finance.

PayLens brings mathematical truth back to multi-rail commerce. Every dollar calculated in code. Every action explained on AMD.